Homework

1. Complete the following.

a. Write in tabular form:

i. $A = \{x \colon x \text{ is a letter in "do not repeat"}\}$

$A = \{d, o, n, t, r, e, p, a\}$

ii. $3\mathbb{Z} = \{x \in \mathbb{Z} \colon x = 3n \text{ for some } n \in \mathbb{Z}\}$

$3\mathbb{Z} = \{\dots, -6, -3, 0, 3, 6, \dots\}$

iii. $N = \{x \colon x \text{ is odd and } x \text{is even}\}$ (recall that 0 is even)

$N = \emptyset$

b. Write these sets in set-builder form:

i. Let $A = \{3, 4, 5, 6, 7, \dots\}$

$A = \{x \in \mathbb{Z} \colon x \geq 3\}$

ii. Let $B = \{3, 5, 7, 9, \dots\}$

$B = \{x \colon x = 2n + 1 \text{ where } n \in \mathbb{Z} \text{ and } n \geq 1\}$

iii. Let $\mathbb{Q}$ be the set of rational numbers.

$\mathbb{Q} = \{x \colon x = \dfrac{a}{b} \text{ where } a, b \in \mathbb{Z}\}$

2. Find all subsets of $A = \{1, 2, 3\}$.

$\{1, 2, 3\}$, $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$, $\{1\}$, $\{2\}$, $\{3\}$, $\emptyset$. There are 3 elements in A so there are $2^3 = 8$ subsets.

3. Prove that every nonempty subset has a proper subset.

Proof. Let A be any non-empty subset. $\emptyset \subseteq A$ since $\emptyset$ is a subset of every set, and $\emptyset \subset A$ (a proper subset) since $A \neq \emptyset$. □

4. Prove that if A is a subset of $\emptyset$, then $A = \emptyset$.

Proof. Let $A \subseteq \emptyset$. $\emptyset \subseteq A$ for any subset A. So then $A = \emptyset$. □

5. Prove that $A = \{2, 3, 4, 5\}$ is not a subset of $B = \{x \mid x \text{ is even}\}$.

Proof. 5 is not even since $5 = 2 \cdot 2 + 1$. So then $5 \notin B$. Thus $A \nsubseteq B$. □

6. Prove that A and B are subsets of $A \cup B$.

Proof. Let $a \in A$. $a \in A$ so then $a \in A \cup B$ by definition of the union of sets. Thus $A \subseteq A \cup B$. Similarly we can show $B \subseteq A \cup B$. □

7. Prove that $A = A \cup A$ and $A = A \cap A$.

Proof. ($A = A \cup A$). By exercise 6, $A \subseteq A \cup A$. So we need only show that $A \cup A \subseteq A$. Let $a \in A \cup A$. Then $a \in A$ or $a \in A$, implying that $A \cup A \subseteq A$. Thus $A \cup A = A$

($A = A \cap A$). Let $a \in A$. Then $a \in A$ and $a \in A$. Thus $A \subseteq A \cap A$. Let $a \in A \cap A$. Then $a \in A$ and $a \in A$ by definition, and this implies that $A \cap A \subseteq A$.

Thus $A = A \cap A$. □

8. Prove that $A \cap B$ is a subset of A and of B.

Proof. Let $x \in A \cap B$. Then $x \in A$ and $x \in B$ by definition. So then $x \in A$ implying that $A \subseteq A \cap B$. Similarly we can show that $B \subseteq A \cap B$. □

9. Prove that $A \cup B = \emptyset$ implies that $A = \emptyset$ and $B = \emptyset$.

Proof. Suppose that $A \cup B = \emptyset$. $A \subseteq A \cup B = \emptyset$ by exercise 6. Thus $A \subseteq \emptyset$, and this implies that $A = \emptyset$ by exercise 4.

Similarly we can show that $B = \emptyset$. □

10. Prove that $A \cup B = B$ if and only if $A \subseteq B$.

Proof. ($\Rightarrow$) Let $A \cup B = B$. $A \subseteq A \cup B = B$ (by exercise 6) which implies that $A \subseteq B$.
($\Leftarrow$) Let $A \subseteq B$. $B \subseteq A \cup B$ (again exercise 6) so we only need to show that $A \cup B \subseteq B$. Let $a \in A \cup B$ then $a \in A$ or $a \in B$. If $a \in B$ then $A \cup B \subseteq B$ as required. If $a \in A$ then $A \subseteq B$ implies that $a \in B$. Thus $A \cup B \subseteq B$. Which implies that $A \cup B = B$. □

- For "if and only if/ $\iff$ " statements, you need to show both directions, $\Rightarrow$ and $\Leftarrow$.
- For $A = B$ statements, it is usually easiest to show that $A \subseteq B$ and $B \subseteq A$ separately.

11. Prove that $A \cap B = B$ if and only if $B \subseteq A$.

Proof. ($\Rightarrow$) Let $A \cap B = B$. $B = A \cap B \subseteq A$ by exercise 8 so then $B \subseteq A$.
($\Leftarrow$) Let $B \subseteq A$. We already know that $A \cap B \subseteq B$ be ex. 8. So we need only show that $B \subseteq A \cap B$. Let $b \in B$. By assumption $B \subseteq A$ so then $b \in A$ which implies that b is in A and B, i.e., $b \in A \cap B$. □

12. Prove that $A \cup (A \cap B) = A$.

Proof. $A \cup (A \cap B) = A$ if and only if $A \cap B \subseteq A$ by ex. 10, and $A \cap B \subseteq A$ by ex. 8. □

13. Prove that $A \cap (A \cup B) = A$.

Proof. $A \cap (A \cup B) = A$ if and only if $A \subseteq A \cup B$ by ex. 11, and $A \subseteq A \cup B$ by ex. 6. □

14. Prove that $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Proof. Let $P \in \mathcal{P}(A \cap B)$. Then $P \subseteq A \cap B$ implying that $P \subseteq A$ and $P \subseteq B$ thus $P \in \mathcal{P}(A) \cap \mathcal{P}(B)$. Now let $P \in \mathcal{P}(A) \cap \mathcal{P}(B)$. Thus $P \subseteq A$ and $P \subseteq B$ which means that $P \subseteq A \cap B$ and that $P \in \mathcal{P}(A \cap B)$. □

Presentation problems.

1. Prove that $A - B$ is a subset of $A \cup B$.

2. Prove that if A a subset of B then $A \cap B = A$.

3. Let $A \cap B = \emptyset$. Prove that $B \cap A' = B$.

4. Prove that if A is a subset of B then $A \cup B = B$.

5. Prove that $A' - B' = B - A$.

6. Prove that if A is a subset of B then B' is a subset of A'.

7. Prove that if $A \cap B = \emptyset$ then $A \cup B' = B'$.

8. Prove that $(A \cap B)' = A' \cup B'$.

9. Prove that $A \subset B$ implies that $A \cup (B - A) = B$.